

Analyzing the Transmission of Monetary Policy Tightening in the Euro Area

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September 25, 2026

Main Focus – Questions

Najjar and Shapiro (2026) modeled the transmission of monetary policy tightening based on decomposing inflation into supply- and demand driven (Shapiro (2025)).

Goal: Create a framework for the implementation of a similar model for the Eurozone

Model and Indicators

$$f_t^{(h)} - f_{t-1}^{(h)} = \alpha + \beta^0 MPS_t + \beta^d \pi_{d,m(t)} \cdot MPS_t + \beta^s \pi_{s,m(t)} \cdot MPS_t + \gamma^d \pi_{d,m(t)} + \gamma^s \pi_{s,m(t)} + \varepsilon_t$$

$f_t^{(h)} - f_{t-1}^{(h)}$ is the difference in interest rate expectation for horizon h , between before and after monetary policy announcement at time t

MPS_t measures/indicates occurrence of a monetary policy surprise

$\pi_{s,m(t)}$ and $\pi_{d,m(t)}$ represent supply- and demand-driven inflation, respectively (Najjar and Shapiro (2026)).

Model and Indicators

Selecting $h = 1$ Year, I use the *1-year (AAA) instantaneous forward rate* for $f_t^{(h)}$ at time t .

For the inflation decomposition, I use data from Fagandini et al. (2024)

Definition of MPS_t derived from Najjar and Shapiro (2026) and Jarociński and Karadi (2020)

Model Results

model_1.0

Regression Results

Variable	Coefficient	Std. Error	t-value	p-value	95% CI
Intercept	-0.0056	0.0009	-6.461	<0.001	[-0.0072, -0.0039]
MPT	-0.0208	0.0115	-1.812	0.070	[-0.0433, 0.0017]
Demand-driven	0.0098	0.0021	4.623	<0.001	[0.0056, 0.0139]
MPT × Demand-driven	0.1346	0.0328	4.098	<0.001	[0.0702, 0.1990]
Supply-driven	0.0039	0.0019	2.092	0.037	[0.0002, 0.0075]
MPT × Supply-driven	-0.0554	0.0305	-1.818	0.069	[-0.1152, 0.0044]

Model Statistics

Observations: 2069 R²: 0.0679 Adjusted R²: 0.0656 F-statistic: 30.0592 F p-value: <0.001

Model Results

Demand-driven inflation exerts a statistically **larger main effect** on interest rate expectation changes **than supply-driven shocks**.

Demand-driven inflation **amplifies** the **effect** size of a **monetary policy tightening** shock on rate expectations, whereas supply shocks do not.

Model Limitations

Caveats:

Indicators only limited proxies of those used by Najjar and Shapiro (2026) (e.g. the AAA Instantaneous 1-year Forward Rate is not exactly analogous to the U.S. indicator used in the original model)

Currently used indicators do not allow more granular intra-day observations

Next step: compare the model's performance to the performance of one that only uses HICPX without decomposition

Tool allows the inclusion of other indicators from the ECB Data Portal and low-effort addition of new API sources: can seamlessly implement models with higher quality variables

Small modifications of the model will allow the inclusion of more granular data

References



Fagandini, Bruno, Eduardo Goncalves, Ieva Rubene, Omiros Kouvavas, Katalin Bodnar, and Gerrit Koester. 2024. "Decomposing HICPX inflation into energy-sensitive and wage-sensitive items." Issue 3/2024, *European Central Bank Economic Bulletin*, https://www.ecb.europa.eu/press/economic-bulletin/focus/2024/html/ecb.ebbox202403_06~bf8222a3ae.en.html.



Jarociński, Marek, and Peter Karadi. 2020. "Deconstructing Monetary Policy Surprises—The Role of Information Shocks." *American Economic Journal: Macroeconomics* 12 (2): 1–43. <https://doi.org/10.1257/mac.20180090>.



Najjar, Rami, and Adam Shapiro. 2026. "Not All Inflation Is the Same: State-Dependent Transmission of Monetary Policy." Working Paper 2025-28, *Federal Reserve Bank of San Francisco*, <https://doi.org/10.24148/wp2025-28>.